

Table of content

Preamble..... 2

Mult.asm program..... 3

Fill.asm program..... 3

Preamble

The present *readme* file gives some informations upon the *Mult* and *Fill* assembler programs available in the [04](#) Bitbucket directory.

Mult.asm program

The code is based on a simple algorithm described in the [asm](#) source file.

Fill.asm program

The program fills the screen memory zone with the content of the [wordvalue](#) variable. Its content is set to 0 to clear the screen or to -1 (thus all bits set to 1) to blacken it. Remembering how to test it was not so easy and since we have to check visually if it runs well, here is how I do it on my Windows PC :

- Call the [CPUEmulator.bat](#) tool.
- *File / Load Program*
- Select [Fill.asm](#) and click *Load ROM*
- *Animate : No animation* and *View : Screen*
- Run (▶▶)
- Check that the keyboard button is selected.
- Maintain one key pressed to blacken the screen, release the key to clear it.

CPU Emulator (2.5) - C:\Perso\Work\nand2tetris\projects\04\fill\Fill.asm

File View Run Help

The screenshot shows the CPU Emulator (2.5) interface. The title bar reads "CPU Emulator (2.5) - C:\Perso\Work\nand2tetris\projects\04\fill\Fill.asm". The menu bar includes "File", "View", "Run", and "Help". The toolbar contains icons for file operations and execution controls. The main interface is divided into several panels:

- ROM Panel:** A list of memory addresses and their corresponding instructions. The first instruction at address 0 is "D=M".
- RAM Panel:** A list of memory addresses and their corresponding values. The first value at address 0 is 23773.
- Screen Display:** A large black rectangle representing the screen memory zone.
- Keyboard Input:** A small keyboard icon with a blue highlight on the 'D' key.
- D Register:** A text box showing the value 0.
- ALU Unit:** A diagram showing the ALU input (D Input: 0, M/A Input: 0) and the ALU output (0).

At the bottom of the interface, a status bar indicates "Running..."